Technical Appendix

mvp20 evidence pack Customer: a top U.S. Navy prime

1. Candidate feasibility bands and cards

Feasibility band	Concepts	Score	Meaning
Band A	C01, C02, C03, C04	0.805	score 0.805; C01-C04; indistinguishable by current model; bench decides; no pot-life/corrosion/spray rank authority
Band B	C05, C06, C09, C10	0.803	score 0.803; C05/C06/C09/C10; minor sourcing/score difference only; bench decides; no pot-life/corrosion/spray rank authority
Band C	C07, C08	0.795	score 0.795; C07/C08; lower sourcing robustness; bench decides; no pot-life/corrosion/spray rank authority

These are feasibility bands, not a performance rank. Pot-life, corrosion, and sprayability have no validated rank authority — the bench decides ordering.

Display	Concept	Resin / hardener	R	Solvents	Extender package	ZnP	Status	VS/VOC/WPG	PVC/Λ/mix
1	C01 dgeba_828 + anca...	dgeba_828 / anca...	0.70	naphtha/n_butanol	A silica 22 phr; B silica 10 phr; additives ...	N	primary	65.0/281/9.16	9.3/0.17/1.00
2	C02 dgeba_828 + anca...	dgeba_828 / anca...	0.70	naphtha/n_butanol	A silica 22 phr; B silica 10 phr; additives ...	N	discovery-adjacent	65.0/281/9.05	8.8/0.16/1.00
3	C03 dgeba_828 + anca...	dgeba_828 / anca...	0.70	naphtha/n_butanol	A silica 28 phr; B silica 12 phr; additives ...	N	discovery-adjacent	65.0/281/9.34	11.0/0.20/1.00
4	C04 dgeba_331 + anca...	dgeba_331 / anca...	0.60	naphtha/n_butanol	A silica 28 phr; B silica 12 phr; additives ...	N	discovery-adjacent	65.0/282/9.39	11.7/0.22/1.00
5	C05 dgeba_331 + anca...	dgeba_331 / anca...	0.60	mineral_spirits/n_butanol	A talc 18 phr; A talc 6 phr; B silica 9 phr; ...	N	discovery-adjacent	65.0/282/9.15	9.4/0.17/1.00
6	C06 dgeba_331 + anca...	dgeba_331 / anca...	0.60	mineral_spirits/n_butanol	A talc 18 phr; A talc 6 phr; B silica 9 phr; ...	N	discovery-adjacent	65.0/282/9.27	9.6/0.18/1.00
7	C07 dgeba_828 + anca...	dgeba_828 / anca...	0.50	mineral_spirits/n_butanol	A talc 5 phr; A silica 20 phr; B silica 10 phr; ...	N	discovery-adjacent	65.0/277/9.02	8.2/0.15/1.00
8	C08 dgeba_331 + anca...	dgeba_331 / anca...	0.50	mineral_spirits/n_butanol	A talc 5 phr; A silica 20 phr; B silica 10 phr; ...	N	discovery-adjacent	65.0/277/9.01	8.2/0.15/1.00
9	C09 dgeba_828 + anca...	dgeba_828 / anca...	0.70	naphtha/n_butanol	A silica 22 phr; B silica 10 phr; additives ...	Y	discovery-adjacent	65.0/282/9.35	10.9/0.20/1.00
10	C10 dgeba_828 + anca...	dgeba_828 / anca...	0.70	naphtha/n_butanol	A silica 22 phr; B silica 10 phr; additives ...	Y	discovery-adjacent	65.0/281/9.23	10.3/0.19/1.00

#	Model outputs	Score components	Red-team gap	CRO validation step
1	solv 0.65/government_precedent; KU 86; barrier 0.463; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.25	The card can still fail by pot-life collapse, phase split, poor spray atomization, weak adhesion, or measured ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...
2	solv 0.65/government_precedent; KU 85; barrier 0.454; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.25	The card can still fail by pot-life collapse, phase split, poor spray atomization, weak adhesion, or measured ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...
3	solv 0.65/government_precedent; KU 87; barrier 0.490; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.25	The card can still fail by pot-life collapse, phase split, poor spray atomization, weak adhesion, or measured ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...
4	solv 0.65/government_precedent; KU 88; barrier 0.457; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.25	The card can still fail by pot-life collapse, phase split, poor spray atomization, weak adhesion, or measured ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...
5	solv 0.65/government_precedent; KU 86; barrier 0.419; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.23	Talc/silica packing could fail if CPVC/oil-absorption estimates miss real wetting, viscosity, or settling ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...
6	solv 0.65/government_precedent; KU 86; barrier 0.424; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.23	Talc/silica packing could fail if CPVC/oil-absorption estimates miss real wetting, viscosity, or settling ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...
7	solv 0.65/government_precedent; KU 85; barrier 0.451; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.15	Mica-containing barrier logic could fail by dispersion, viscosity/sag, or opacity problems before corrosion ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...
8	solv 0.65/government_precedent; KU 85; barrier 0.451; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.15	Mica-containing barrier logic could fail by dispersion, viscosity/sag, or opacity problems before corrosion ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...
9	solv 0.65/government_precedent; KU 87; barrier 0.476; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.23	Functional zinc phosphate could fail by hydrophilic blistering, inhibitor-dose non-monotonicity, supplier ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...

10	solv 0.65/government_precedent; KU 87; barrier 0.466; cure FAILED/DEMOTED model card; corrosion ...	lambda 1.00; precedent 1.00; evap 0.50; sourcing 0.23	Functional zinc phosphate could fail by hydrophilic blistering, inhibitor-dose non-monotonicity, supplier ...	Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, ...

C01 dgeba_828 + ancamide_350a R=0.7 naphtha/n_butanol no-ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band A: score 0.805; C01-C04; indistinguishable by current model. bench decides; no pot-life/corrosion/spray rank authority.
Metric strip: VS 65.0%; VOC 281 g/L; WPG 9.16; PVC/lambda 9.3/0.17; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 86; cure FAILED/DEMOTED model card; barrier 0.463; corrosion FAILED/DEMOTED; sourcing 0.25
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.25; feasibility score 0.805
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

C02 dgeba_828 + ancamide_260a R=0.7 naphtha/n_butanol no-ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band A: score 0.805; C01-C04; indistinguishable by current model. bench decides; no pot-life/corrosion/spray rank authority.
Metric strip: VS 65.0%; VOC 281 g/L; WPG 9.05; PVC/lambda 8.8/0.16; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 85; cure FAILED/DEMOTED model card; barrier 0.454; corrosion FAILED/DEMOTED; sourcing 0.25
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.25; feasibility score 0.805
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

C03 dgeba_828 + ancamide_2353 R=0.7 naphtha/n_butanol no-ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band A: score 0.805; C01-C04; indistinguishable by current model. bench decides; no pot-life/corrosion/spray rank authority.
Metric strip: VS 65.0%; VOC 281 g/L; WPG 9.34; PVC/lambda 11.0/0.20; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 87; cure FAILED/DEMOTED model card; barrier 0.490; corrosion FAILED/DEMOTED; sourcing 0.25
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.25; feasibility score 0.805
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

C04 dgeba_331 + ancamide_350a R=0.6 naphtha/n_butanol no-ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band A: score 0.805; C01-C04; indistinguishable by current model. bench decides; no pot-life/corrosion/spray rank authority.
Metric strip: VS 65.0%; VOC 282 g/L; WPG 9.39; PVC/lambda 11.7/0.22; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 88; cure FAILED/DEMOTED model card; barrier 0.457; corrosion FAILED/DEMOTED; sourcing 0.25
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.25; feasibility score 0.805
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

C05 dgeba_331 + ancamide_260a R=0.6 mineral_spirits/n_butanol no-ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band B: score 0.803; C05/C06/C09/C10; minor sourcing/score difference only. bench decides; no pot-life/corrosion/spray rank ...
Metric strip: VS 65.0%; VOC 282 g/L; WPG 9.15; PVC/lambda 9.4/0.17; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 86; cure FAILED/DEMOTED model card; barrier 0.419; corrosion FAILED/DEMOTED; sourcing 0.23
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.23; feasibility score 0.803
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

C06 dgeba_331 + ancamide_2353 R=0.6 mineral_spirits/n_butanol no-ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band B: score 0.803; C05/C06/C09/C10; minor sourcing/score difference only. bench decides; no pot-life/corrosion/spray rank ...
Metric strip: VS 65.0%; VOC 282 g/L; WPG 9.27; PVC/lambda 9.6/0.18; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 86; cure FAILED/DEMOTED model card; barrier 0.424; corrosion FAILED/DEMOTED; sourcing 0.23
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.23; feasibility score 0.803
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

C07 dgeba_828 + ancamide_220x70 R=0.5 mineral_spirits/n_butanol no-ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band C: score 0.795; C07/C08; lower sourcing robustness. bench decides; no pot-life/corrosion/spray rank authority.
Metric strip: VS 65.0%; VOC 277 g/L; WPG 9.02; PVC/lambda 8.2/0.15; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 85; cure FAILED/DEMOTED model card; barrier 0.451; corrosion FAILED/DEMOTED; sourcing 0.15
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.15; feasibility score 0.795
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

C08 dgeba_331 + ancamide_220x70 R=0.5 mineral_spirits/n_butanol no-ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band C: score 0.795; C07/C08; lower sourcing robustness. bench decides; no pot-life/corrosion/spray rank authority.
Metric strip: VS 65.0%; VOC 277 g/L; WPG 9.01; PVC/lambda 8.2/0.15; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 85; cure FAILED/DEMOTED model card; barrier 0.451; corrosion FAILED/DEMOTED; sourcing 0.15
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.15; feasibility score 0.795
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

C09 dgeba_828 + ancamide_350a R=0.7 naphtha/n_butanol ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band B: score 0.803; C05/C06/C09/C10; minor sourcing/score difference only. bench decides; no pot-life/corrosion/spray rank ...
Metric strip: VS 65.0%; VOC 282 g/L; WPG 9.35; PVC/lambda 10.9/0.20; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 87; cure FAILED/DEMOTED model card; barrier 0.476; corrosion FAILED/DEMOTED; sourcing 0.23
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.23; feasibility score 0.803
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

C10 dgeba_828 + ancamide_260a R=0.7 naphtha/n_butanol ZnP
Badges: UNKNOWN CANDIDATE, NO WET-LAB CLAIM
Feasibility band B: score 0.803; C05/C06/C09/C10; minor sourcing/score difference only. bench decides; no pot-life/corrosion/spray rank ...
Metric strip: VS 65.0%; VOC 281 g/L; WPG 9.23; PVC/lambda 10.3/0.19; mix 1.00; coverage/WFT 209/7.7
Model strip: solvency 0.65/government_precedent; KU 87; cure FAILED/DEMOTED model card; barrier 0.466; corrosion FAILED/DEMOTED; sourcing 0.23
Score components -> lambda 1.00; precedent 1.00; solvency 0.65; evap 0.50; sourcing 0.23; feasibility score 0.803
CRO validation step: Run 250 g compatibility/KU screen, 1 kg/component drawdowns, Method 24/D2697/D1475, adhesion, B117/D5894/D870/ISO 16773 EIS ...

2. Known-control recovery checks

Display	Check	P/F	Evidence	Validates	Does not validate
1	F160 black public formula	PASS	KNOWN FULL-BOM	formula parsing; density/VOC envelope arithmetic; known-control sanity ...	top recovery proof; candidate performance; corrosion performance; pot ...
2	F161 yellow public formula	PASS	KNOWN FULL-BOM	formula parsing; density/VOC envelope arithmetic; known-control sanity ...	top recovery proof; candidate performance; corrosion performance; pot ...
3	Evonik SPF A702B75P1	PASS	KNOWN FULL-BOM	formula parsing; density/VOC envelope arithmetic; known-control sanity ...	top recovery proof; candidate performance; corrosion performance; pot ...
4	NCP MIL-DTL-24441 Type IV anchor	PASS	KNOWN PARTIAL-TDS	class-envelope recovery; 1:1 epoxy-polyamide precedent check	full-BOM reconstruction; corrosion performance; pot life; ...
5	International Formula 150 anchor	PASS	KNOWN PARTIAL-TDS	class-envelope recovery; 1:1 epoxy-polyamide precedent check	full-BOM reconstruction; corrosion performance; pot life; ...
6	over-pigmented (talc 300 phr)	PASS	NEGATIVE CONTROL	fail-closed gate behavior; negative-control rejection	candidate performance; candidate corrosion; candidate pot life; ...
7	stoichiometry R=1.3	PASS	NEGATIVE CONTROL	fail-closed gate behavior; negative-control rejection	candidate performance; candidate corrosion; candidate pot life; ...
8	80% volume-solids demand	PASS	NEGATIVE CONTROL	fail-closed gate behavior; negative-control rejection	candidate performance; candidate corrosion; candidate pot life; ...

F160 black public formula
Badges: PASS
Display 1; pass/fail PASS; KNOWN FULL-BOM
Expected behavior: {"behavior": "recompute available full-BOM arithmetic and disclose residuals", "credit_limit": "formula parsing / densit
Engine result residuals: {"mix_ratio_delta_from_1_to_1": -0.024, "voc_margin_g_l": 55.293, "volume_solids_pts": 3.844}
Validates: formula parsing; density/VOC envelope arithmetic; known-control sanity check
Does not validate: top recovery proof; candidate performance; corrosion performance
Caveat: Full-BOM recovery uses public formula constants and class substitutions; it is not authority authorization, corrosion proof, or a ...

F161 yellow public formula
Badges: PASS
Display 2; pass/fail PASS; KNOWN FULL-BOM
Expected behavior: {"behavior": "recompute available full-BOM arithmetic and disclose residuals", "credit_limit": "formula parsing / densit
Engine result residuals: {"mix_ratio_delta_from_1_to_1": -0.03, "voc_margin_g_l": 56.178, "volume_solids_pts": 3.956}
Validates: formula parsing; density/VOC envelope arithmetic; known-control sanity check
Does not validate: top recovery proof; candidate performance; corrosion performance
Caveat: Full-BOM recovery uses public formula constants and class substitutions; it is not authority authorization, corrosion proof, or a ...

Evonik SPF A702B75P1
Badges: PASS
Display 3; pass/fail PASS; KNOWN FULL-BOM
Expected behavior: {"behavior": "recompute available full-BOM arithmetic and disclose residuals", "credit_limit": "formula parsing / densit
Engine result residuals: {"density_g_ml": 0.038, "voc_g_l": 14.0, "vs_pts": -1.31}
Validates: formula parsing; density/VOC envelope arithmetic; known-control sanity check
Does not validate: top recovery proof; candidate performance; corrosion performance
Caveat: Trade-name additive/filler lines use flagged approximate class constants; this is a post-hoc check and never a candidate ranking input.

NCP MIL-DTL-24441 Type IV anchor
Badges: PASS
Display 4; pass/fail PASS; KNOWN PARTIAL-TDS
Expected behavior: {"behavior": "recover the known epoxy-polyamide class envelope only", "credit_limit": "partial-TDS anchor; no full-BOM r
Engine result residuals: {"mix_ratio_delta_from_reported": 0.0, "voc_margin_g_l": 54.501, "volume_solids_pts": -1.5}
Validates: class-envelope recovery; 1:1 epoxy-polyamide precedent check
Does not validate: full-BOM reconstruction; corrosion performance; pot life
Caveat: Partial-TDS anchors support class recovery only; they cannot be treated as full formulation reconstruction or corrosion superiority ...

International Formula 150 anchor
Badges: PASS
Display 5; pass/fail PASS; KNOWN PARTIAL-TDS
Expected behavior: {"behavior": "recover the known epoxy-polyamide class envelope only", "credit_limit": "partial-TDS anchor; no full-BOM r
Engine result residuals: {"mix_ratio_delta_from_reported": 0.0, "voc_margin_g_l": 48.501, "volume_solids_pts": 4.0}
Validates: class-envelope recovery; 1:1 epoxy-polyamide precedent check
Does not validate: full-BOM reconstruction; corrosion performance; pot life
Caveat: Partial-TDS anchors support class recovery only; they cannot be treated as full formulation reconstruction or corrosion superiority ...

over-pigmented (talc 300 phr)
Badges: PASS
Display 6; pass/fail PASS; NEGATIVE CONTROL
Expected behavior: {"behavior": "reject deliberately bad formulation by the registered gate", "expected_gate": "lambda_below_cpvc/pvc_windo
Engine result residuals: {"computed_failed_gates": ["lambda_below_cpvc", "pvc_window", "weight_per_gal"], "correct_expected_gate_hit": true}
Validates: fail-closed gate behavior; negative-control rejection
Does not validate: candidate performance; candidate corrosion; candidate pot life
Caveat: A negative control proving rejection is a guardrail check, not evidence that any surviving candidate performs in a wet lab.

stoichiometry R=1.3
Badges: PASS
Display 7; pass/fail PASS; NEGATIVE CONTROL
Expected behavior: {"behavior": "reject deliberately bad formulation by the registered gate", "expected_gate": "stoichiometry"}
Engine result residuals: {"computed_failed_gates": ["pvc_window", "stoichiometry"], "correct_expected_gate_hit": true}
Validates: fail-closed gate behavior; negative-control rejection
Does not validate: candidate performance; candidate corrosion; candidate pot life
Caveat: A negative control proving rejection is a guardrail check, not evidence that any surviving candidate performs in a wet lab.

80% volume-solids demand
Badges: PASS
Display 8; pass/fail PASS; NEGATIVE CONTROL
Expected behavior: {"behavior": "reject deliberately bad formulation by the registered gate", "expected_gate": "volume_solids"}
Engine result residuals: {"coverage", "volume_solids", "wft"], "correct_expected_gate_hit": true}
Validates: fail-closed gate behavior; negative-control rejection
Does not validate: candidate performance; candidate corrosion; candidate pot life
Caveat: A negative control proving rejection is a guardrail check, not evidence that any surviving candidate performs in a wet lab.

3. Envelope-65 CRO recipe card

Var	Part	Material	Role	g	wt%	250g/1kg/5gal g	Tolerance

black	part_A	dgeba_828	DGEBA epoxy binder	100.00	72.78	104.36/727.8/8673	+/-1.0% mass unless CoA value forces recalculation
black	part_A	silica	barrier extender	22.00	16.01	22.96/160.1/1908	+/-2.0% mass; stop if dispersion/QC target misses
black	part_A	dispersant	wetting/dispersing additive	1.20	0.87	1.25/8.7/104	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
black	part_A	defoamer	air-release additive	0.50	0.36	0.52/3.6/43	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
black	part_A	thixotrope	anti-settle/sag rheology ...	1.80	1.31	1.88/13.1/156	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
black	part_A	iron_oxide_black	black identification pigment	3.50	2.55	3.65/25.5/304	+/-2.0% mass; stop if dispersion/QC target misses
black	part_A	naphtha	solvent	8.40	6.11	8.77/61.1/729	+/-1.0% mass unless CoA value forces recalculation
black	part_B	ancamide_350a	polyamide/amidoamine ...	37.13	36.34	38.75/363.4/3220	+/-1.0% mass unless CoA value forces recalculation
black	part_B	silica	barrier extender	10.00	9.79	10.44/97.9/867	+/-2.0% mass; stop if dispersion/QC target misses
black	part_B	dispersant	wetting/dispersing additive	0.80	0.78	0.83/7.8/69	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
black	part_B	thixotrope	anti-settle/sag rheology ...	1.20	1.18	1.25/11.7/104	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
black	part_B	n_butanol	solvent	53.03	51.91	55.34/519.1/4599	+/-1.0% mass unless CoA value forces recalculation
yellow	part_A	dgeba_828	DGEBA epoxy binder	100.00	73.09	104.61/730.9/8673	+/-1.0% mass unless CoA value forces recalculation
yellow	part_A	silica	barrier extender	22.00	16.08	23.01/160.8/1908	+/-2.0% mass; stop if dispersion/QC target misses
yellow	part_A	dispersant	wetting/dispersing additive	1.20	0.88	1.25/8.8/104	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
yellow	part_A	defoamer	air-release additive	0.50	0.36	0.52/3.7/43	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
yellow	part_A	thixotrope	anti-settle/sag rheology ...	1.80	1.32	1.88/13.2/156	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
yellow	part_A	iron_oxide_yellow	yellow identification ...	2.92	2.13	3.06/21.3/253	+/-2.0% mass; stop if dispersion/QC target misses
yellow	part_A	naphtha	solvent	8.40	6.14	8.79/61.4/729	+/-1.0% mass unless CoA value forces recalculation
yellow	part_B	ancamide_350a	polyamide/amidoamine ...	37.13	36.34	38.84/363.4/3220	+/-1.0% mass unless CoA value forces recalculation
yellow	part_B	silica	barrier extender	10.00	9.79	10.46/97.9/867	+/-2.0% mass; stop if dispersion/QC target misses
yellow	part_B	dispersant	wetting/dispersing additive	0.80	0.78	0.84/7.8/69	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
yellow	part_B	thixotrope	anti-settle/sag rheology ...	1.20	1.18	1.25/11.7/104	+/-0.05 g at 250 g scale or +/-5%, whichever is ...
yellow	part_B	n_butanol	solvent	53.03	51.91	55.48/519.1/4599	+/-1.0% mass unless CoA value forces recalculation

Process parameters

{"filtration_packout_retains": "Filter compatible with pigment/additive package; pack Part A and Part B separately; retain sealed samples with lot, fill mass, and date.", "grind_endpoint": "Hegman ≥6 / no hard agglomerates; stop and review if endpoint not reached before temperature limit", "induction_mix_use": "Mix A:B 1:1 by volume, record material temperature, induce 30 min at 73F unless CRO pre-approves deviation, start pot-life clock at first contact.", "letdown_instructions": "Add solvent in two cuts while recording temperature, KU trend, and density; do not adjust outside VOC envelope without engineering signoff.", "max_temperature_c": 55, "mix_speed_shear_target": "Cowles: low-speed wet-out 500-800 rpm, dispersion 1500-2500 rpm or lab-equivalent tip speed; avoid vortex air entrainment", "mix_time": "wet-out 10-15 min; dispersion 20-35 min or until Hegman endpoint; letdown 10 min minimum", "order_of_addition": ["Verify CoA/SDS/TDS and pre-stage lots; stop on missing declaration-critical fields.", "Part A: charge DGEBA resin, dispersant, and defoamer; start low-speed mix.", "Part A: add silica and color pigment slowly under Cowles dispersion; add thixotrope after wet-out.", "Part A: let down with naphtha to target mass/viscosity; hold below max temperature.", "Part B: charge polyamide curative and dispersant; add silica and thixotrope under moderate shear.", "Part B: let down with n-butanol; filter/pack only after density/KU/visual checks."]}

QC targets / stop rules / returned fields

Gate	Method	Target	Fail / inconclusive rule	Raw return field
gel_or_phase_split	visual check plus KU trend during induction and first ...	0 observed gel/phase-split events	repeat/review if control invalid; otherwise stop or reformulate on ...	visual_observation, ku_vs_time_3temps.raw_value, ...
pot_life_below_90_min	ASTM D562 KU-vs-time at 65/73/95F plus application ...	≥90 min at 73F	repeat/review on control failure or noisy tie; stop/reformulate if ...	pot_life_h.raw_value, ku_vs_time_3temps.raw_value, ...
voc_over_340_g_l	EPA Method 24 / ASTM D3960	≤340 g/L	inconclusive if worksheet missing; stop/reformulate if valid value ...	voc_g_l.raw_value, method24_worksheet_path
volume_solids_outside_63_67	ASTM D2697 or lab-accepted Method 24 solids worksheet	63-67%	inconclusive if raw replicates/worksheet missing; stop/reformulate if ...	volume_solids_pct.raw_value, replicate_values, ...
severe_sag_or_pinhole	drawdown + airless spray observation, ASTM D4400/D1210	Hegman ≥6; sag ≥12 mil target	inconclusive if setup missing; stop/reformulate if valid application ...	fineness_hegman.raw_value, sag_mils.raw_value, ...
failed_adhesion	ASTM D4541 dry and post-immersion	≥20% margin vs threshold/control basis where ...	repeat/review on adhesive failure or control failure; stop on valid ...	adhesion_mpa.raw_value, failure_mode, dolly_map_path
early_corrosion_failure	ASTM B117/D1654, ASTM D870, ISO 16773 EIS	no tied/noisy decisive pair; EIS gap target ...	controls fail -> repeat/review, not pass; tied/noisy decisive pair -> ...	eis_immersion.raw_spectra_path, ...

Returned field	Required content
method	string ASTM/ISO/EPA/internal SOP identifier
unit	string unit for the raw value
condition	temperature/RH/substrate/immersion/spray condition
replicate_count	integer n, minimum set by field
deviation_code	one of DEV-NONE/DEV-PREP/DEV-BATCH/DEV-ENV/DEV-APP/DEV-METHOD/DEV-DATA

raw_value	number, string, or structured list of replicate raw values
pass_fail	PASS/FAIL/REVIEW/ABSENT
raw_file_path	machine-readable raw export path when an instrument generates one
attachment_path	photo, EIS file, panel image, raw instrument export, or lab notebook path

4. Full M0–M15 model ledger

ID	Submodel	Status	Data used	Method	Metric/result	Red-team gap	Next closure
M0	Corpus and source ledger	SANITY CHECK	106 source records; 83 compiled constants.	Versioned JSON source pack plus citation ledger compile gate.	source-record schema and citation presence: mvp20.source_records.v4; constants compiled=83	URLs and public extracts do not equal supplier declarations or authority signoff.	Refresh source records and RFQ declarations before bench spend.
M1	Formula reconstruction	ARITHMETIC RECOMPUTED	MIL F160/F161 public sheets, STIC Type IV, F150/NCP anchors.	Ingredient-level mass/volume recompute and architecture checks.	density and architecture recovery checks: F160 density recovery A=8.22 vs reported; ...	Public formula reconstruction is a benchmark, not proof that a new 65% package qualifies.	Keep Lineage-60 as CRO control and compare measured panels side by side.
M2	VOC/solids/density/mix-ratio ...	ARITHMETIC RECOMPUTED	Compiled constants and enumerated candidate recipes.	Closed-form simultaneous solvent solve plus deterministic mixture ...	hard gates and recompute identities: 3975 candidates solved; 2444 hard-gate survivors.	Arithmetic says the numbers close; it does not prove sprayability or corrosion.	Verify by Method 24, D2697, D1475, drawdowns, and CRO batch records.
M3	Arithmetic audit	ARITHMETIC RECOMPUTED	Generated candidate packages and government-lineage controls.	Independent recompute checks over output formulas.	all audit checks pass: 6/6 checks pass across 4 formulas.	Audit catches arithmetic drift, not bad assumptions or bad supplier data.	Add CRO-returned measured values as future audit rows.
M4	Authority and claim guard	SANITY CHECK	Claim registry, QPL/F-718/NESHAP/EPA source records, patent landscape.	Explicit customer-safe claim text and banned-claim lint.	artifact lint and claim traceability: Authority lock carried in PDFs; legal/FTO/procurement ...	String guards cannot replace counsel, contracting, or authority review.	Counsel and contracting authority review after supplier declarations and lab data exist.
M5	Known-material rediscovery	SANITY CHECK	F160/F161, Evonik SPF A702B75P1, F150/NCP Type IV anchors, negative ...	Post-hoc recompute and class-envelope comparison.	registered recompute bars and gate rejection: examples=SPF-A702B75P1, NAVAL-CLASS-RECOVERY, ...	Rediscovery is sanity evidence, not novelty, qualification, or performance proof.	Run Lineage-60 and a commercial reference control if obtainable in the same CRO batch.
M6	Solvency model	LIMITED SANITY	Hansen HSP tables where public; government solvent-precedent fallback ...	Hansen RED arithmetic plus fail-closed precedent fallback.	RED<1 for precedent blends; ordering sanity: verdict=PASS; reds={'xylene_butanol': 0.609, ...	Low-N sanity card; naphtha/mineral spirits rely on precedent, not measured compatibility.	CRO compatibility, blush, phase-split, and pot-life screens.
M7	Viscosity KU proxy	LIMITED SANITY	F160/NCP class KU anchors and lead derived metrics.	Interpretable KU proxy: solids crowding plus pigment-volume term.	predict class anchors inside the sprayable band + correct solids/PVC ordering: ...	Low-shear KU band is not high-shear atomization, sag, or hiding.	D562/D4400 spray/sag map and yellow D2805 contrast-ratio gate.
M8	Cure/pot-life model	NO AUTHORITY	Evonik cure verification set and independent pot-life anchors.	LOOCV gel-time regression; second feature attempt with same frozen ...	LOOCV multiplicative fold-error on gel time: best within x/1.5=0.438; bar=0.80; verdict=FAIL	Cannot rank candidates by pot-life; real solvents, pigment heat, and induction can ...	Week-1 KU-vs-time and usable-pot-life screen before full corrosion battery.
M9	Corrosion rank model	NO AUTHORITY	Open corrosion series and in-class table-read filters.	Pre-registered literature validation bar; failed closed to design ...	Spearman $\geq$0.6 on $\geq$3 independent strictly-IN-CLASS + TABLE-READ barrier-epoxy series: ...	Corrosion rank is unvalidated; no candidate corrosion prediction has rank authority.	Side-by-side B117/D5894/D870/ISO 16773 EIS with locked rank and null baselines.
M10	Barrier tortuosity model	DIRECTIONAL ONLY	Barrier theory, within-study filler/PVC ladders, candidate derived ...	Transparent barrier index; within-study directional checks only.	within-study DIRECTIONAL agreement (never pooled cross-study); pre-registered track rank: ...	Directional model can size a falsification test, not claim measured corrosion.	Register locked rank before panels; compare measured rank against solids-only and ...
M11	Raw EIS pipeline	DIRECTIONAL ONLY	Open CC0 EIS dataset plus synthetic forward examples.	Equivalent-circuit fit and power-analysis scaffold.	raw spectra fit quality and scope label: available=True; n_spectra=12; median R2=0.9926	Off-class single-panel spectra do not rank this chemistry.	Use CRO-returned raw EIS files from this exact formulation family.
M12	Resin/curative property ...	EXACT STRUCTURE ONLY	data/molecule_library.json: 40+ real molecules (SMILES/CAS/PubChem ...	EXACT structure->property (rdkit: EEW=MolWt/epoxide-count, ...	held-out MAE on known equivalent weights + rediscovery: EEW MAE 0.167 g/eq, AHEW MAE 0.033 ...	Exact ONLY for a defined monomer; commercial resins/polyamides are polydisperse so their ...	Train the estimated-tier Tg/viscosity models on CRO-measured + public data; add conformal ...
M13	Full cure kinetics model	ROADMAP	Not yet implemented; requires rheology/time, FTIR/DSC, temperature ...	Roadmap mechanistic/empirical kinetics model.	none yet: No full cure kinetics model exists in mvp20.	Current failed pot-life surrogate proves this cannot be inferred from public gel data ...	Collect KU-vs-time, dry-to-touch, through-cure, and recoat data at 65/73/95F.
M14	Dispersion/manufacturing/QC ...	ROADMAP	Not yet implemented; would require batch records, Hegman, ...	Roadmap process twin plus QC anomaly model.	none yet: No manufacturing digital twin, dispersion model, or QC anomaly model is used.	Process failures can dominate the formulation despite good arithmetic.	Instrument screen/bench/pilot batches and feed deviations back to the twin.
M15	Bayesian CRO optimizer	ROADMAP	Not yet implemented; requires measured CRO outcomes and cost/time ...	Roadmap Bayesian optimizer over formulation and process variables.	none yet: No active-learning optimizer is used; current search is deterministic enumeration ...	Without wet-lab returns, optimization would only optimize surrogate assumptions.	Run first CRO batch, ingest raw outcomes, then let optimizer propose batch 2.

5. Clean-slate provenance

Item	User	Engine	Gen	Validation	Guard
Human technical envelope / problem constraints	Y	N	Y	N	N
F160/F161 public MIL formula lineage	N	Y	N	Y	Y
F150/NCP Type IV partial-TDS anchors	N	Y	N	Y	N
Evonik SPF A702B75P1 starting-point formulation	N	Y	N	Y	N
Supplier raw-material families	N	Y	Y	N	N
Corrosion literature	N	Y	N	N	N
QPL/F-718 authority records	N	Y	N	N	Y

6. Run manifest

Audit artifact	Status	Detail
Whole-run output hash (sha256)	PRESENT	e15c20df360d12cd18e42888...
Code tree hash (sha256)	PRESENT	31b73a3b148091cc28c77226...
Git commit	UNAVAILABLE	Command '['git', 'rev-parse', 'H...
Frozen scoring weights	PRESENT	published in funnel + manifest
Gate/scoring config versions	PRESENT	mvp20.gates.v1+cro_capture.v2 / mvp20.funnel_weights.v1
MC seed / source-query hash	PRESENT	2001 / 74509b975ab9293e...
Barrier locked-rank hash	RECOMPUTES	197cf16aa46bca218a53...
Input-PDF sha256 (x4)	PRESENT	government slash sheets
No manual candidate edits after run start	ATTESTED	deterministic enumeration; byte-identical reruns
Third-party pre-registration	REQUIRED	before panels are cut

Engine-proof signal	Current mvp20 evidence
Corpus compiled	106 source records; 83 constants audited
Candidate count	4320 enumerated; 3975 solved onto the arithmetic surface
Gate survivors	2444 hard-gate survivors; 3 Monte-Carlo robust carry tracks
Top chemistry cards	10 de-duplicated raw-material formulation chemistry concepts
Known-material rediscovery	F160/F161 reconstruction; Evonik SPF A702B75P1 recompute; F150/NCP Type IV class anchors; negative controls rejected
Autonomy boundary	no humans in the computational selection loop; lab, EHS, legal/FTO, and procurement approval remain human-controlled

7. Source quality and source table

KPI	Current value
Constants total	83
Two-source constants	11
Arithmetic-crosschecked constants	8
Single-source constants	48
Manual-required value comparisons	49
Load-bearing violations	0
Confidence	arithmetic HIGH; source values MEDIUM; sourcing LOW

Source	Kind	Status	Revision / safe use
SRC-QPL-24441	authority	resolved / HIGH	last update reported by source: 27-JAN-2026 | official authority read / verify at use | QPL-24441 is the procurement source of truth; page reports MIL-DTL-24441D and ...
SRC-MIL-DTL-24441D	authority	snapshot_only / HIGH	MIL-DTL-24441D public copy; verify amendments before procurement use | not numeric / bibliographic anchor | Governing public specification for epoxy-polyamide paints; qualification still ...
SRC-F160	authority	snapshot_only / HIGH	slash sheet /38A public copy | not numeric / bibliographic anchor | Public black Type-IV lineage and government formula benchmark.
SRC-F161	authority	snapshot_only / HIGH	slash sheet /39B public copy | not numeric / bibliographic anchor | Public yellow Type-IV lineage and color-twin benchmark.
SRC-F718	authority	resolved / HIGH	NAVSEA-reviewed sheet; verify current sheet before shipboard use | not numeric / bibliographic anchor | F-718 review covers application process only and does not qualify a material or ...
SRC-EPA-M24	authority	resolved / HIGH	EPA method page current as retrieved; confirm method revision before certification | not numeric / bibliographic anchor | VOC, volatile matter, density, volume solids and weight solids method anchor.
SRC-SSPC-SP10	authority	paywalled / MEDIUM	standard page; full standard is controlled | not numeric / bibliographic anchor | Panel-preparation anchor for near-white blast; execution details must come from ...
SRC-OSHA-1910-106	authority	resolved / HIGH	current CFR web page as retrieved | not numeric / bibliographic anchor | Flammable-liquid handling control anchor for solvent storage, transfer, ...
SRC-NFPA-30	authority	blocked / MEDIUM	code landing page; edition must be selected for facility design | not numeric / bibliographic anchor | EHS design anchor for flammable-liquid storage/handling; not a substitute for ...
SRC-SCAQMD-1106	authority	resolved / MEDIUM	district rule PDF; verify local rule at point of use | not numeric / bibliographic anchor | Local VOC-rule caution: stricter district/state limits may override the ...
SRC-EPON828	supplier	resolved / HIGH	TDS file path dated 2024-06; supplier revision must be requested | unknown / RFQ required | EEW/density source for a property-window line, not a brand lock.

SRC-EVONIK-GUIDE	supplier	resolved / MEDIUM	public guide; request current TDS/SDS by grade before purchase | unknown / RFQ required | AHEW/density class windows and alternate curative candidates.
SRC-EASTMAN-NBUTANOL	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | primary and second source read / RFQ required | Solvent property anchor; HAP/TSCA/PFAS/declarations still required.
SRC-CAMEO-SILICA	supplier	resolved / MEDIUM	public chemical record; supplier grade declarations still required | unknown / RFQ required | Mineral density anchor for silica extender; not a commercial grade approval.
SRC-BYK-ADDITIVES	supplier	resolved / LOW	dynamic product page; request current TDS/SDS | unknown / RFQ required | Small additive property-window example; exact grade must be selected by CRO ...
SRC-ELEMENTIS-BENTONE	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | RFQ/declaration required | Rheology-additive class anchor; exact grade requires RFQ and IP review.
SRC-WO2016187380	patent	resolved / MEDIUM	patent publication; legal status must be verified by counsel | not numeric / bibliographic anchor | Claim-chart comparison only; not an FTO opinion.
SRC-EP3328938	patent	resolved / MEDIUM	patent publication; legal status must be verified by counsel | not numeric / bibliographic anchor | Claim-chart comparison only; not an FTO opinion.
SRC-US5059640	patent	resolved / MEDIUM	patent publication; legal status must be verified by counsel | not numeric / bibliographic anchor | Expired Navy-assigned prior art; claim chart only.
SRC-WO2013159277	patent	resolved / MEDIUM	patent publication; legal status must be verified by counsel | not numeric / bibliographic anchor | Overcoatability claim-family anchor for absence of chlorinated polyolefin in ...
SRC-EP2931822	patent	resolved / LOW	patent publication; legal status must be verified by counsel | not numeric / bibliographic anchor | Epoxy blend claim-family anchor for absence of CHDM epoxy in the candidate.
SRC-NPJ-2024	literature	resolved / MEDIUM	2024 article; verify corrections before modeling use | not numeric / bibliographic anchor | Literature support for active-learning roadmap only; no performance claim for ...
SRC-BO-ML-2025	literature	paywalled / LOW	2025 article landing page; full text access not verified | not numeric / bibliographic anchor | Candidate reference for future closed-loop optimization; not load-bearing in ...
SRC-CPVC-EIS	literature	paywalled / LOW	article landing page; full text access not verified | not numeric / bibliographic anchor | Candidate CPVC/PVC support; paywalled/abstract-only in this pass, not used as a ...
SRC-TSCA-INVENTORY	authority	resolved / HIGH	EPA page last updated 10-JUN-2026; inventory download must be checked by CAS/PMN at purchase | not numeric / bibliographic anchor | TSCA inventory is a regulatory lookup starting point; individual BOM CAS ...
SRC-NESHAP-SHIPYARD	authority	resolved / HIGH	eCFR current page as retrieved; official annual CFR may lag current eCFR | not numeric / bibliographic anchor | Shipyard surface-coating NESHAP applicability and low-usage exemptions are ...
SRC-ASTM-D2697	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Volume-solids method anchor for CRO; no controlled text reproduced.
SRC-ASTM-D1475	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Density cup method anchor for WPG verification.
SRC-ASTM-B117	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Salt-fog exposure anchor; not a standalone performance claim.
SRC-ASTM-D5894	authority	paywalled / MEDIUM	controlled ASTM method page; buy/read current standard for execution | not numeric / bibliographic anchor | Cyclic corrosion arm is comparative; raw cycle logs and panel images are ...
SRC-ASTM-D870	authority	paywalled / MEDIUM	controlled ASTM method page; buy/read current standard for execution | not numeric / bibliographic anchor | Immersion arm is the primary falsification test context for rank ordering.
SRC-ASTM-D4541	authority	paywalled / MEDIUM	controlled ASTM method page; buy/read current standard for execution | not numeric / bibliographic anchor | Adhesion is never predicted; CRO returns raw force/failure mode and photos.
SRC-ASTM-D2805	authority	paywalled / MEDIUM	controlled ASTM method page; buy/read current standard for execution | not numeric / bibliographic anchor | Yellow opacity/contrast ratio is a bench gate, not an arithmetic color-swap ...
SRC-ISO-16773	authority	paywalled / MEDIUM	controlled ISO standard page; buy/read current standard for execution | not numeric / bibliographic anchor | EIS arm requires raw spectra/time series; cross-study EIS values are not pooled.
SRC-NIST-SDL	literature	resolved / HIGH	published 07-SEP-2024; NIST page updated 11-JUL-2025 | not numeric / bibliographic anchor | Supports the next-engine roadmap: autonomous experimentation, data standards, ...
SRC-CAMEO-NAPHTHA	supplier	resolved / MEDIUM	public chemical record; supplier grade declarations still required | unknown / RFQ required | Naphtha solvent property anchor; procurement still needs SDS/TDS, ...
SRC-LANXESS-BLACK-OXIDE	supplier	resolved / MEDIUM	public TDS; request current SDS/TDS and declaration before purchase | primary read only / RFQ required | Black iron oxide candidate grade; density/oil absorption are property-window ...
SRC-LANXESS-YELLOW-OXIDE	supplier	resolved / MEDIUM	dynamic product page; request current TDS/SDS and declaration before purchase | unknown / RFQ required | Yellow iron oxide candidate grade; opacity/contrast ratio remains a CRO bench ...
SRC-ZINC-PHOSPHATE-ZP10	supplier	resolved / MEDIUM	Rev. 01.02 / 01-2022 public TDS; request current SDS/TDS and declaration before purchase | primary read only / RFQ required | Zinc phosphate is functional optional and must be corrosion-justified before ...
SRC-ECFR-63-II	authority	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Applies to shipbuilding and ship repair surface coating operations at major ...
SRC-ASTM-D2369	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Weight solids/volatile method anchor used by Method 24 workflows.

SRC-ASTM-D3960	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | VOC calculation/reporting method anchor; Method 24 measured result governs.
SRC-ASTM-D562	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | KU viscosity method anchor for pot-life and sprayability screen.
SRC-ASTM-D1210	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Hegman grind endpoint anchor.
SRC-ASTM-D4400	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Sag resistance method anchor; execution by CRO SOP.
SRC-ASTM-D1654	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Scribe/field rating anchor after corrosion exposure.
SRC-ASTM-D1849	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Accelerated storage anchor; 36-month shelf claim remains extrapolation.
SRC-ASTM-D869	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Settling evaluation anchor for retained Part A/Part B packages.
SRC-ASTM-D4417	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Blast profile measurement anchor.
SRC-ASTM-D4414	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | WFT measurement anchor for spray/drawdown panels.
SRC-ASTM-D3278	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Flash-point method anchor for slash-sheet checks.
SRC-ASTM-D3828	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | F-718 flash point method anchor.
SRC-SSPC-PA2	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | DFT grid/inspection anchor; CRO SOP and current standard govern.
SRC-ISO-8502-9	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Soluble-salt method anchor; not reproduced.
SRC-NACE-SP0508	authority	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Soluble-salt equivalence anchor; F-718 example cites this family.
SRC-NFPA-33	authority	blocked / MEDIUM	current landing page as retrieved; verify current revision before use | not numeric / bibliographic anchor | Spray booth/application EHS anchor; code text controlled.
SRC-EPA-TSCA-INVENTORY	authority	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | RFQ/declaration required | TSCA inventory status must be supplier-declared by CAS and grade; this page is ...
SRC-EPA-PFAS-REPORTING	authority	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | RFQ/declaration required | PFAS reporting/declaration anchor; no BOM line can be closed without supplier ...
SRC-FAR-25-4	authority	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | RFQ/declaration required | TAA/domestic-origin screen anchor; contracting counsel and supplier COO ...
SRC-CA-PROP65	authority	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | RFQ/declaration required | Prop65 declaration anchor; supplier declaration required for each grade.
SRC-ECHA-REACH	authority	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | RFQ/declaration required | REACH screen anchor; supplier declaration required.
SRC-EPON828-TDS	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | primary read plus ledger second source / CoA required / RFQ required | Candidate resin property-window anchor; buy only after CoA and declarations.
SRC-DER331-TDS	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | primary read plus ledger second source / CoA required / RFQ required | Alternate liquid DGEBA resin anchor; declarations unknown.
SRC-YD128-TDS	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | single-source waiver in value audit / RFQ required | Alternate resin anchor only; source and declarations require RFQ.
SRC-ANCAMIDE-350A	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | primary and guide read / CoA required / RFQ required | Candidate curative property-window anchor; declarations unknown.
SRC-EVONIK-CURING-GUIDE	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | guide read / RFQ required | Guide anchor for curative family windows; individual grade declarations ...
SRC-VERSAMID-115	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | waived single source in value audit / RFQ required | Lineage curative anchor; ownership and current availability must be ...
SRC-DOW-NBUTANOL	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | secondary numeric read / RFQ required | Second-source solvent property anchor; declarations unknown.
SRC-SILICA-CAMEO	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | handbook read / RFQ required | Mineral density anchor; supplier SDS/CoA required for grade-specific hazards ...
SRC-BYK-DISPERBYK-163	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | RFQ/declaration required | Additive-class anchor; exact grade must be screened and supplier-declared.
SRC-BYK-A530	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | RFQ/declaration required | Additive-class anchor; exact grade must be screened and supplier-declared.

SRC-TALC-303	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | waived single-source / RFQ required | Alternate extender anchor; grade-specific declarations required.
SRC-MICA-N	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | primary read only / RFQ required | Alternate lamellar-extender anchor; declarations required.
SRC-WOLLASTONITE-M400	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | primary read only / RFQ required | Alternate extender anchor; declarations required.
SRC-CARBON-BLACK	supplier	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | estimate / RFQ required | Small-loading colorant anchor; exact grade and declarations required.
SRC-CORR-ASPERTI-2024	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | literature table read / no performance claim | Open table-read epoxy-amine nanoclay ladder; used only for within-study ...
SRC-CORR-ZHU-2022	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | literature table read / no performance claim | Open table-read epoxy-amine graphene ladder; used only for within-study ...
SRC-CORR-RAMESH-2012	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | figure/page read / no performance claim | Near-class PVC ladder used to support low-PVC/CPVC design rule only.
SRC-CORR-CAI-2021	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | literature read / not rank validation | Closest chemistry class but only a modification comparison; not an ordered ...
SRC-CORR-NIKRAVESH-PVC	literature	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | controlled/paywalled bibliographic anchor | On-topic PVC/CPVC bibliographic anchor; numeric values not used.
SRC-CORR-CLAY-REVIEW	literature	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | controlled/paywalled bibliographic anchor | Near-class nanoclay barrier anchor; no numeric rank validation.
SRC-CORR-SELF-HEAL-REVIEW	literature	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | controlled/paywalled bibliographic anchor | Discovery-annex anchor only; not in baseline formula.
SRC-WATERBORNE-EPOXY-LIT	literature	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | controlled/paywalled bibliographic anchor | Adjacent chemistry only; not part of solvent-borne envelope.
SRC-PAT-WATERBORNE-EPOXY	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-ZINC-RICH	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-GRAPHENE-EPOXY	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-NANOCLAY-EPOXY	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-SELF-HEALING	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-MICROCAPSULE	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-PHENALKAMINE	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-AMIDOAMINE	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-POLYSILOXANE	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-BIO-EPOXY	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-MOLYBDATE	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-PAT-PROCESS-QC	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-EP2653485	patent	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | not FTO / counsel review required | Google Patents category-search anchor only; not a family-status or FTO ...
SRC-CORR-NIKRAVESH-MIO-AL	literature	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | abstract extraction / no strict numeric table read | Closest binder/curative lamellar-pigment target; public preview supports CRO ...
SRC-CORR-RODRIGUEZ-CPVC	literature	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | public preview extraction / no full metric table read | Epoxy primer Lambda ladder with public levels and conclusion; not polyamide and ...
SRC-CORR-NADERI-ZAPP-CD	literature	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | abstract extraction / no side-by-side numeric table read | On-topic phosphate/cathodic-disbonding anchor; public preview supports gate ...
SRC-CORR-KAPANI-ZP	literature	paywalled / MEDIUM	current landing page as retrieved; verify current revision before use | public numeric highlight / not baseline candidate validation | Open preview contains a long-duration EIS highlight and formulation materials; ...

SRC-CORR-MANCHESTER-SAPP	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | open PDF extraction / adjacent system | Open PDF explains phosphate/polyphosphate inhibition and zinc-phosphate ...
SRC-COMM-CARBOLINE-F160	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | reference control unknown / RFQ required | Public product-page candidate only; exact lot, QPL listing, SDS/TDS revision, ...
SRC-COMM-CARBOLINE-F160-PDS	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | reference control unknown / RFQ required | Public PDS candidate only; verify current revision and acquire lot-specific kit ...
SRC-COMM-SHERWIN-TYPEIV	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | reference control unknown / RFQ required | Public product-line candidate only; exact product/color/lot and current ...
SRC-COMM-NCP-TYPEIV	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | reference control unknown / RFQ required | Public TDS-library candidate only; exact product/lot/QPL ...
SRC-COMM-GRACO-NCP	literature	resolved / MEDIUM	current landing page as retrieved; verify current revision before use | reference control unknown / RFQ required | Public distributor candidate only; listing requires related specification ...

8. Known materials and references

F160/F161 reconstruction: component densities recompute to A 8.22 and B 10.24 lb/gal inside the public reported windows. Evonik SPF A702B75P1 recompute: VS 60.7% vs reported 62.0%, density 1.448 vs 1.41 g/mL, VOC 334 vs 320 g/L. F150/NCP Type IV class anchors: lead VS 65.0% / VOC ∼281 g/L sits inside the customer envelope; public F160/F161 black/yellow remain at 60.2+/-2% solids, a major lineage-vs-envelope variance. Negative controls rejected: 3/3.

Claim	Tier	Customer-safe wording	Sources
CL-AUTH-001	AUTHORITY	Candidate only: not QPL-listed, not NAVSEA-qualified, and not an approval for procurement.	SRC-QPL-24441, SRC-F718
CL-AUTH-002	AUTHORITY	Qualification status must be verified against QPL-24441; the DLA page retrieved 2026-07-03 reports last update 27-JAN-2026.	SRC-QPL-24441
CL-LINEAGE-001	AUTHORITY	The 65% black/yellow package is envelope-literal and materially different from qualified public Formula 160/161.	SRC-F160, SRC-F161
CL-FORM-001	REAL-ARITHMETIC	Envelope-65 is the primary phosphate-free candidate; color pigments are identification only.	SRC-EPA-M24
CL-FORM-002	REAL-ARITHMETIC	Enhanced-65 must be justified by corrosion testing before zinc phosphate is retained.	SRC-US5059640
CL-NUM-001	REAL-ARITHMETIC	Black/yellow twins are equal-volume color swaps and remain within the arithmetic parity bars.	SRC-EPA-M24
CL-NUM-002	REAL-ARITHMETIC	WPG, PVC/CPVC, and stoichiometry are computed fields, not supplier or model claims.	SRC-EPA-M24
CL-NUM-003	REAL-ARITHMETIC	Film-build and coverage numbers are arithmetic targets that still require CRO drawdown/spray confirmation.	SRC-EPA-M24
CL-WET-001	ABSENT	Pot life, corrosion, adhesion, sprayability, sag and shelf life remain CRO bench gates.	none
CL-CORR-001	MODELED-AND-FAILED	Corrosion is not predicted; B117/D5894/EIS/immersion testing decides.	SRC-NPJ-2024, SRC-CPVC-EIS
CL-CURE-001	MODELED-AND-FAILED	Pot life is not predicted; the week-1 KU-vs-time screen is mandatory.	none
CL-CRO-001	TEST-TARGET	Batch sizes are execution gates, not proof of scale-up success.	none
CL-CRO-002	TEST-TARGET	Surface preparation and environment are required returned-data fields.	SRC-SSPC-SP10
CL-CRO-003	TEST-TARGET	Failed early gates stop or reformulate before paid scale-up.	SRC-EPA-M24
CL-IP-001	LEGAL-SCOPE	Not an FTO opinion; counsel must review claims, jurisdictions, prosecution history and equivalents.	SRC-WO2016187380, SRC-EP3328938, SRC-US5059640, SRC-WO2013159277, SRC-EP2931822
CL-SUPPLY-001	SUPPLIER-DOC	Sourcing is RFQ-gated; domestic/TAA and PFAS/TSCA claims require supplier declarations.	SRC-EPON828, SRC-EVONIK-GUIDE, SRC-EASTMAN-NBUTANOL, SRC-CAMEO-SILICA, SRC-BYK-ADDITIVES, SRC-ELEMENTIS-BENTONE
CL-SUPPLY-002	UNKNOWN	Supplier declaration fields are rendered as unknown / RFQ required unless verified.	SRC-EPON828, SRC-EVONIK-GUIDE, SRC-BYK-ADDITIVES, SRC-ELEMENTIS-BENTONE
CL-ENGINE-001	ROADMAP	Current capability is deterministic constraint solving plus validation-card discipline; active-learning autonomy is the next engine.	SRC-NPJ-2024, SRC-BO-ML-2025, SRC-NIST-SDL
CL-FALS-001	TEST-TARGET	The locked barrier-rank protocol can lose; it is registered before panels are cut and compared against hostile nulls.	none
CL-CRO-004	TEST-TARGET	CRO execution requires controls, randomization, raw files, calibration records, and minimum panel counts.	SRC-ASTM-B117, SRC-ASTM-D5894, SRC-ASTM-D870, SRC-ASTM-D4541, SRC-ISO-16773

CL-SUPPLY-003	UNKNOWN	Supplier PFAS/TSCA/TAA/origin/lead-time fields stay unknown / RFQ required until declaration evidence is attached.	SRC-TSCA-INVENTORY, SRC-EPON828, SRC-EVONIK-GUIDE, SRC-CAMEO-NAPHTHA, SRC-LANXESS-BLACK-OXIDE, SRC-LANXESS-YELLOW-OXIDE, SRC-ZINC-PHOSPHATE-ZP10

9. Corrosion theory and literature

10. Corrosion: a cited mechanistic barrier model (honest state)

Corrosion is the mission property, and it is honestly the weakest link – no wet-lab data on our own tracks exists. What Phase 4 adds is a *cited, mechanistic* barrier-permeability model (Nielsen 1967 / Cussler 1988 platelet tortuosity with an aggregation term), replacing the earlier heuristic index, fit *within each measured study only* (cross-study EIS is non-comparable, so no value is ever pooled). Two prior attack surfaces are now **answered from the literature**: (i) the EIS bridge $\log_{10} R_c = \log_{10} R_{c,0} - \log_{10} P_r$ is the classical **Maxwell-Fricke effective-medium equivalence** – the same geometric tortuosity τ/ϵ sets both effective permeability and effective conductivity, so $R_c/R_{c,0} = \sigma_0/\sigma_{eff} = 1/P_r$ (MacMullin-Muccini 1956; Fricke 1924; Landesfeind 2016; already used in coating EIS by Duarte 2009, Deflorian 2006, Policastro 2023) – valid in the intact/high-$|Z|$ regime, and it bridges R_c ↔permeability, NOT R_c ↔corrosion-rate (Sykes 2017); (ii) the 2–5 decade magnitude that *dilute Nielsen* cannot reach IS reached by Cussler ($\alpha^2\phi^2$) / Gusev-Lusti (exponential), which we now fit and select by AIC. The one remaining modeled closure we flag is the aggregation kernel $\alpha_{eff} = \alpha_0/(1+k\phi)$ – a mechanistically-motivated (tactoid-stacking α/N) but not named-literature form, *built* to allow a peak.

A reproducible pipeline demo on real data — with its limits stated: On the open CC0 dataset (Zenodo 4985780), the standard circuit R_s-p(R_{po},CPE) fits real EIS spectra and recovers a measured ∼**40.2**× R_{po} improvement for the platelet coating. But honestly: this is **one panel vs one panel** (not replicated); a 4-parameter circuit fits almost any smooth $|Z|$, so the high R^2 oversells; and on this **weak, off-class** (waterborne, $|Z|$ ∼10^5–10^6) coating, R_{po} is defect/pore-dominated — i.e. it measures a mechanism the tortuosity model does *not* capture. It shows the pipeline runs; it is NOT an in-class validated prediction.

Where the model is honestly still weaker: even with the right models and the cited bridge, the *empirical fit* to the sparse open in-class ladders (4–5 pts, dispersion outliers, |Z| swinging into the defect regime) stays weak in both timepoints – best in-class R^2 ≈0.6 with α railed. We fit the intact/high-$|Z|$ regime per Maxwell-Fricke and still do **not** count optimum-reproduction (kernel-placed) as corroboration. The equiaxed ZnO/polyamide ladder is correctly a poor fit – a genuine mechanism control, though that contrast is a cross-chemistry qualitative argument, not a within-study fit.

Ladder	Class	Read	best R^2	What the model gets
asperti2024-nanoclay	epoxy-amine	TABLE	+0.32	magnitude under-predicted
zhu2022-graphene	epoxy-amine	TABLE	+0.64	optimum + direction
go-amine-2022	epoxy-amine (waterborn	FIGURE (approx)	-1.31	magnitude under-predicted
zno-polyamide-2020	epoxy-POLYAMIDE (polya	TABLE	-0.31	optimum + direction
graphene-wb-epoxy-zenodo	waterborne epoxy (hard	TABLE + DOWNLOADABLE raw spectra	+0.90	optimum + direction

Blunt scoring: the framework is now **well-cited end-to-end** (bridge + magnitude objections answered) – it has earned the right to stop being called bookkeeping. But it is honestly **6–7/10**, not 8: there is still **zero in-class, high-$|Z|$ validated barrier prediction** because the *data* does not exist openly – the exact solvent-borne epoxy-polyamide/MIO ladders (Corros. Sci. 2011; IJES 2023) are paywalled, the Manchester 500-spectra set is undeposited, and the best open in-class ladders (nanoclay 5e9; GO 1e11) are too sparse for a strong fit. A defensible 8 needs one in-class, high-$|Z|$ ($\geq 10^9$) dataset whose barrier factor matches a *pre-registered* prediction in our chemistry – retrieved via an institutional library, or the founders' own ≥6-panel bench (the falsification test). The physics also sizes that bench. A Fickian → Brasher-Kingsbury → equivalent-circuit forward model (design-only; every reported number still comes from panels) gives a standard power analysis: with panel scatter $\sigma \approx 0.2$–0.5 decade of $\log|Z|$ (a literature prior, *not* yet measured on our panels), **6 panels/track/arm** resolves a 0.5–1 decade barrier gap at 80% power ($n = \lceil (z_\alpha + z_\beta)^2\, 2\sigma^2/\Delta^2 \rceil$), EIS timepoints at days 0/1/3/7/14/28/60.

Study	Class	Variable	Read	Use limit

asperti2024	epoxy-amine	nanoclay Cloisite 30B wt%	table	optimum at 1 wt%; 3% dip from clay clustering (dispersion-limited)
zhu2022	epoxy-amine	graphene wt% in fixed 35% mica/epoxy	table	optimum at 0.7 wt%; over-loading at 1.0% degrades barrier
ramesh2012	epoxy-ester (near-class; PVC ladder)	PVC %	figure	MONOTONIC: lower PVC = better barrier – directly supports the lambda<CPVC / low-PVC design rule
cai2021	epoxy-POLYAMIDE (E44+TY-650)	mica modification (unmod vs mechanochem 200C)	figure	closest to our epoxy-polyamide chemistry but only a 2-level modification comparison (not a content ...
nikravesh2005_pvc_epoxy	epoxy-amine (paywalled PVC/CPVC ladder)	PVC/CPVC ratio	figure	On-topic PVC/CPVC anchor; excluded from model validation because numeric table was not open-read.
mio_ladder_2023	epoxy-amine (MIO lamellar pigment)	MIO wt% ladder	figure	On-topic lamellar pigment ladder; not open table-read in this pass.
mica_ladder_2023	epoxy-amine (mica lamellar pigment)	mica wt% ladder	figure	On-topic lamellar pigment ladder; not used as validation because numeric table was not open-read.
zinc_phosphate_epoxy_anchor	epoxy-amine (inhibitor package)	zinc phosphate inhibitor package	figure	Supports why Enhanced-65 is a testable inhibitor variant; does not validate performance.
graphene_go_epoxy_anchor	epoxy-amine (graphene/GO nanofiller)	graphene oxide loading/surface treatment	figure	Discovery-annex nanofiller evidence; not part of baseline.
nanoclay_epoxy_anchor	epoxy-amine (nanoclay filler)	nanoclay wt% and dispersion	figure	Supports non-monotonic filler caveat; exact values not used.
self_healing_review_2018	adjacent epoxy self-healing	self-healing chemistry category	figure	Discovery-annex only; no self-healing additive is in Envelope-65.
npj_ml_self_healing_2024	adjacent epoxy self-healing / ML	ML-designed self-healing epoxy formulation space	figure	Roadmap anchor for future active learning; not evidence for current candidate.
waterborne_epoxy_adjacent	adjacent waterborne epoxy	waterborne epoxy dispersion and cure package	figure	Outside solvent-borne customer envelope.
zinc_rich_epoxy_adjacent	adjacent zinc-rich epoxy	metallic zinc loading / cathodic network	figure	Different mechanism and weight target; not part of barrier-primer package.
immersion_adhesion_epoxy_pa	epoxy-polyamide / immersion adhesion	immersion adhesion/cure variables	figure	Explains why adhesion/immersion stay CRO gates.
rodriguez2004_cpvc	epoxy coating (PVC/CPVC ladder)	PVC/CPVC ratio	figure	Strong design-rule anchor for PVC/CPVC sensitivity; not enough to rank candidates.
nikravesh2011_mio_pa	epoxy-polyamide (MIO/Al lamellar pigment)	MIO/Al pigment ratio	figure	Highest-value strict target for MIO/Al lamellar effects; useful for CRO arms, not prediction.
nobakht2022_cathodic_disbonding	epoxy-polyamide cathodic disbonding	DFT, applied potential, immersion time, NaCl concentration, temperature, pH	figure	Directly supports cathodic-disbonding/adhesion as a CRO gate.
khan2023_zp_epoxy	epoxy coating with zinc phosphate (non-polyamide / Al alloy)	neat epoxy vs zinc-phosphate-modified epoxy	table	Zinc-phosphate hypothesis anchor only; not a basis for Enhanced-65 performance claim.
shelf_stability_d1849_pack	epoxy shelf stability / settling	package stability and settling	figure	Shelf life remains an acceptance gate, not a corrosion evidence upgrade.
naderi2010_zapp_epoxy	epoxy coating with ZAPP modified zinc phosphate	no pigment vs zinc phosphate vs zinc aluminum polyphosphate	figure	Useful for Enhanced-65 falsification framing, not candidate performance.

ramezanzadeh2022_epoxy_zinc	epoxy primer with Ka@PAni	zinc phosphate ZP vs Ka/ZP vs Ka@PAni/ZP	figure	Moves CRO design and falsification framing; does not predict Enhanced-65.
liu_manchester_sapp_phos	polyester coil primer with SAPP phosphate pigment	strontium aluminium polyphosphate primer degradation/inhibition	figure	Useful inhibitor-mechanism context; not a candidate ranking record.

11. Patent / FTO landscape

Not an FTO opinion; counsel must review claims, jurisdictions, prosecution history and equivalents.

Area	Anchor	Claim elements	Our package	Read / scope
high-solids epoxy/polyamide	WO2016187380A1	>70% volume solids; polyamide hardener; reactive compatibilizer	Envelope-65 is ∼65% solids and has no separate reactive compatibilizer	UNCLEAR / not FTO / ceased / published family screen / not FTO; claim-1 screen only; NO not assigned because no independent claim chart is attached
reactive compatibilizers	WO2016187380A1	distinct reactive polar compatibilizer	no distinct compatibilizer package	UNCLEAR / not FTO / ceased / published family screen / not FTO; dependent claims and equivalents unread; NO not assigned because no independent claim chart is attached
polyamide rheology additives	EP3328938A1/B1	specific micronized polyamide powder chemistry; 1-10 um particle size	uses a conventional organoclay-class thixotrope, not the claimed micronized polyamide powder	UNCLEAR / not FTO / active/granted B1 in EP family; counsel review required / not FTO; live-grant counsel review required if a polyamide-powder additive is selected; NO not assigned because no independent claim chart is attached
rheology additives	EP3328938A1/B1	specified diamine/monoacid/fatty-acid powder	CRO package keeps rheology additives as supplier-selected non-polyamide alternatives unless screened	UNCLEAR / not FTO / active/granted B1 in EP family; counsel review required / not FTO; category row retained for legacy test coverage; NO not assigned because no independent claim chart is attached
anticorrosive inhibitors	US5059640A	zinc phosphate; zinc molybdate; zinc benzoate triad	primary has no zinc phosphate; Enhanced-65 has zinc phosphate only, not the full triad	UNCLEAR / not FTO / expired US Navy prior art, legal status still counsel-check / not FTO; expired-prior-art status still needs counsel confirmation; NO not assigned because no independent claim chart is attached
anticorrosive packages	US5059640A	complete inhibitor triad	Enhanced-65 is not the triad and is functional optional only	UNCLEAR / not FTO / expired US Navy prior art, legal status still counsel-check / not FTO; legacy category name retained; NO not assigned because no independent claim chart is attached
overcoatability	WO2013159277A1	chlorinated polyolefin terpolymer	no chlorinated polyolefin terpolymer	UNCLEAR / not FTO / ceased / published family screen / not FTO; only anchor family screened; NO not assigned because no independent claim chart is attached

nanofillers	npj 2024 epoxy corrosion ML and public graphene/nanoclay literature	nano-additive corrosion/barrier enhancement; dispersion sensitivity	no graphene, nanoclay, MOF, or nano self-healing package in Envelope-65	OUT OF BASELINE / status unknown / counsel check required / not FTO; discovery-annex only
self-healing systems	npj 2024 epoxy corrosion ML	encapsulated or responsive healing agents; EIS/corrosion outcome data	no self-healing additives in Envelope-65 or Enhanced-65	OUT OF BASELINE / status unknown / counsel check required / not FTO; roadmap/literature only
public MIL prior art	MIL-DTL-24441/38A and /39B	public epoxy-polyamide lineage; black/yellow slash sheets	commodity architecture deliberately stays close to public lineage while solving a 65% envelope	PUBLIC PRIOR ART / public government prior art / not FTO; public-source prior-art posture only
expired Navy prior art	US5059640A	expired Navy-assigned anticorrosive epoxy patent	Enhanced-65 adds zinc phosphate only and does not use the full triad	PUBLIC PRIOR ART / expired US Navy prior art, legal status still counsel-check / not FTO; legal status and term require counsel check
Navy/public-domain prior art	MIL-DTL-24441/38A, /39B and US5059640A	public epoxy-polyamide lineage; expired Navy-assigned anticorrosive patent	commodity architecture deliberately stays close to public prior art	PUBLIC PRIOR ART / expired US Navy prior art, legal status still counsel-check / not FTO; legacy category name retained
adjacent waterborne epoxy systems	public waterborne epoxy literature	waterborne dispersion; coalescence/cure package; water-management constraints	Envelope-65 is solvent-borne and not a waterborne dispersion	OUT OF BASELINE / roadmap category, no clearance / not FTO; discovery-annex only
zinc-rich systems	public zinc-rich primer prior art	metallic zinc dust at cathodic-protection loading; conductive pigment network	no zinc dust and no zinc-rich cathodic-protection mechanism	OUT OF BASELINE / adjacent prior-art category; counsel check required if pursued / not FTO; adjacent system only
process/QC trade-secret space	batch records, CRO residual dataset, source graph	process windows; data feedback loop; claim/source traceability	current protectable asset is process/data discipline, not commodity formula claims	MOAT CATEGORY / not patent clearance / not FTO; business/IP strategy only
WO2016187380A1	WO2016187380A1	epoxy; polyamide hardener; reactive polar compatibilizer; VS >70%; blush rating 4-5	no distinct reactive polar compatibilizer; our VS ∼65% (<70%). Two independent limitations missing.	UNCLEAR / not FTO / Ceased / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
EP3328938B1	EP3328938B1	specified diamine; straight-chain monoacid; specific fatty acid; narrow molar ratio; 1-10 um powder	claims a specific micronized polyamide RHEOLOGY additive, not a curing agent; our dimer-acid liquid polyamide curative is different chemistry. The ONLY live-granted family here.	UNCLEAR / not FTO / Active/granted (2023-05-17; est. expiry ∼2036) / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
EP2931822	EP2931822 via US2015/0329739A1 family member (EP claims did not render)	blend of ≥2 epoxies; one is a CHDM epoxy; CHDM epoxy <2 wt% chlorine	we use standard DGEBA only, no CHDM epoxy.	NO (caveat: read US family claim, not EP) / EP shown Active; US member Ceased / not FTO; category/family screen only
WO2013159277A1	WO2013159277A1	epoxy; chlorinated polyolefin terpolymer; curing agent	no chlorinated polyolefin terpolymer in our primer.	UNCLEAR / not FTO / Ceased / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
US5059640A	US5059640A	epoxy base; Zn phosphate; Zn molybdate; Zn salt of benzoic acid (all three)	primary has no zinc phosphate; Enhanced-65 has zinc phosphate alone, with no molybdate/benzoate; also EXPIRED.	UNCLEAR / not FTO / Expired (∼2003, fee lapse) / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached

waterborne epoxy systems	Google Patents category screen: waterborne epoxy systems	waterborne dispersion; emulsifier/coalescent; aqueous cure package	Envelope-65 is solvent-borne and does not use waterborne dispersion architecture.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
zinc-rich primers	Google Patents category screen: zinc-rich primers	metallic zinc dust; conductive pigment network; cathodic protection	No zinc dust or cathodic-protection pigment network is present.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
graphene epoxy anticorrosion	Google Patents category screen: graphene epoxy anticorrosion	graphene or graphene oxide; dispersion/surface treatment; barrier enhancement	No graphene-family nanofiller is present in Envelope-65 or Enhanced-65.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
nanoclay epoxy barrier	Google Patents category screen: nanoclay epoxy barrier	organoclay/nanoclay; surface modifier; barrier tortuosity	The BOM uses an organoclay rheology additive at low level, not a claimed nanoclay barrier package; exact grade needs review.	UNCLEAR / category screen; individual family status unverified / not FTO; category/family screen only
microcapsule self-healing epoxy	Google Patents category screen: microcapsule self-healing epoxy	microcapsules; healing agent or inhibitor; triggered release	No microcapsules or self-healing package are present.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
intrinsic self-healing epoxy	Google Patents category screen: intrinsic self-healing epoxy	reversible covalent/noncovalent bonds; self-healing matrix	Commodity DGEBA/polyamide matrix lacks designed reversible self-healing chemistry.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
phenalkamine epoxy curatives	Google Patents category screen: phenalkamine epoxy curatives	phenalkamine curative; low-temperature cure	Baseline uses dimer-acid polyamide/amidoamine curative, not phenalkamine.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
amidoamine epoxy curatives	Google Patents category screen: amidoamine epoxy curatives	specified amidoamine adduct; cure accelerator; amine value window	Candidate curative is commodity class, but exact selected grade needs supplier/IP screen.	UNCLEAR / category screen; individual family status unverified / not FTO; category/family screen only
polysiloxane epoxy hybrids	Google Patents category screen: polysiloxane epoxy hybrids	polysiloxane resin; epoxy hybrid network	No polysiloxane or hybrid network is present.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached

bio-based epoxy coatings	Google Patents category screen: bio-based epoxy coatings	bio-based epoxy resin; renewable feedstock	Baseline uses commodity petroleum DGEBA-class epoxy.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
phosphate/molybdate inhibitor packages	Google Patents category screen: phosphate/molybdate inhibitor packages	phosphate inhibitor; molybdate; organic carboxylate salt	Enhanced-65 has zinc phosphate only; no molybdate/benzoate triad, but inhibitor substitutions need review.	UNCLEAR / category screen; individual family status unverified / not FTO; category/family screen only
corrosion indicator smart coatings	Google Patents category screen: corrosion indicator smart coatings	corrosion indicator; responsive additive	No indicator chemistry is present.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
high-solids epoxy process/QC	Google Patents category screen: high-solids epoxy process/QC	process sequence; dispersion/QC window; high solids	Formula is commodity, but process/QC implementation can still create FTO/trade-secret questions.	UNCLEAR / category screen; individual family status unverified / not FTO; category/family screen only
reactive diluent epoxy coatings	Google Patents category screen: reactive diluent epoxy coatings	reactive diluent; low VOC/high solids	No reactive diluent is used in Envelope-65.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
CHDM epoxy blend adjacencies	Google Patents category screen: CHDM epoxy blend adjacencies	CHDM epoxy; epoxy blend; low viscosity	No CHDM epoxy resin is in the BOM.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
chlorinated polyolefin tiecoats	Google Patents category screen: chlorinated polyolefin tiecoats	chlorinated polyolefin; overcoatability	No chlorinated polyolefin terpolymer is present.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
layered MIO barrier pigments	Google Patents category screen: layered MIO barrier pigments	MIO lamellar pigment; barrier pigment loading	No MIO package is included in Envelope-65.	UNCLEAR / not FTO / category screen; individual family status unverified / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached
hydrophobic silica epoxy barriers	Google Patents category screen: hydrophobic silica epoxy barriers	hydrophobic silica; surface treatment; barrier property	Baseline uses silica extender; surface-treated specialty silica would need grade-level review.	UNCLEAR / category screen; individual family status unverified / not FTO; category/family screen only
marine epoxy primer public prior art	Google Patents category screen: marine epoxy primer public prior art	public formula lineage; epoxy-polyamide primer	This is prior-art posture, not an exclusionary claim against the candidate.	UNCLEAR / not FTO / public prior-art category screen / not FTO; category/family screen only; NO not assigned because no independent claim chart is attached

accelerated polymer automation/process twin	Google Patents category screen: accelerated polymer automation/process twin	automation; optimization; formulation data loop	Current mvp20 is deterministic and evidence-controlled, not yet a full active-learning lab engine.	UNCLEAR / category screen; individual family status unverified / not FTO; category/family screen only
EP2653485A2	EP2653485A2	epoxy primer; polyurea topcoat; fast cure; low VOC coating system	baseline is a standalone epoxy-polyamide primer decision package, not a primer/polyurea topcoat system	NO / public bibliographic status only; counsel verification required / not FTO; category/family screen only
WO2009019296A1	WO2009019296A1	tetra-alkoxyorthosilicate reaction product; epoxy resin with hydroxyl groups; $\geq$70 vol% solids or $\leq$250 g/L VOC	baseline does not intentionally use silicate-reacted epoxy, but process chemistry and equivalents are not charted enough for clearance	UNCLEAR / not FTO / public bibliographic status only; counsel verification required / not FTO; category/family screen only
WO2014093115A2	WO2014093115A2	CHDM epoxy resin; high solids; low viscosity epoxy blend	baseline DGEBA resin excludes CHDM by design, but exact family status and claim scope require counsel	UNCLEAR / not FTO / public bibliographic status only; counsel verification required / not FTO; category/family screen only
WO2018010615A1	WO2018010615A1	polyamide rheology modifier; nonionic hydrophilic segment; aqueous coating	baseline is solvent-borne, but additive-grade selection still needs supplier-specific IP review	UNCLEAR / not FTO / public bibliographic status only; counsel verification required / not FTO; category/family screen only
US4865653A	US4865653A	zinc phosphate conversion bath; hydroxylamine agent; metal pretreatment	conversion pretreatment is outside the BOM, but surface-prep choices remain customer/lab scoped	UNCLEAR / not FTO / expired family likely but counsel must verify / not FTO; category/family screen only
US3178319A	US3178319A	phosphate bath; metal conversion coating; salt spray improvement	pretreatment chemistry is outside the candidate paint formulation	UNCLEAR / not FTO / expired family likely but counsel must verify / not FTO; category/family screen only
EP0398202A1	EP0398202A1	zinc phosphate layer; process additive; metal pretreatment	surface pretreatment, not in-can primer composition	UNCLEAR / not FTO / public bibliographic status only; counsel verification required / not FTO; category/family screen only
EP1865013B1	EP1865013B1	polyamide curing agent; dimer acid; polyethyleneamine; epoxy curing	candidate uses commodity polyamide class but grade-specific curative IP is unresolved	UNCLEAR / not FTO / public bibliographic status only; counsel verification required / not FTO; category/family screen only
US6391464B1	US6391464B1	epoxy-coated hull; epoxy-silicone adhesive tiecoat; foul-release overcoat	overcoat/tiecoat architecture is adjacent to but not part of the baseline primer	UNCLEAR / not FTO / public bibliographic status only; counsel verification required / not FTO; category/family screen only
US2705223A	US2705223A	epoxy resin; polyamide resin; thermosetting composition	public prior-art posture for epoxy/polyamide architecture; not a clearance opinion	UNCLEAR / not FTO / expired family likely but counsel must verify / not FTO; category/family screen only
US2986539A	US2986539A	specific polyamide resin; epoxy resin; adhesive composition	adjacent prior-art family; not a coating-performance clearance	UNCLEAR / not FTO / expired family likely but counsel must verify / not FTO; category/family screen only
C09D163/00 sweep family 01 - solvent-borne epoxy anticorrosive primer	Google Patents public family screen: C09D163/00 sweep family 01 - solvent-borne epoxy anticorrosive primer	solvent-borne epoxy anticorrosive primer; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only

C09D163/00 sweep family 02 - bisphenol-A/F epoxy blend coating	Google Patents public family screen: C09D163/00 sweep family 02 - bisphenol-A/F epoxy blend coating	DGEBA/DGEBF blend high-solids coating; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D163/00 sweep family 03 - epoxy phenalkamine marine coating	Google Patents public family screen: C09D163/00 sweep family 03 - epoxy phenalkamine marine coating	phenalkamine-cured marine epoxy; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D163/00 sweep family 04 - epoxy amidoamine corrosion primer	Google Patents public family screen: C09D163/00 sweep family 04 - epoxy amidoamine corrosion primer	amidoamine-cured corrosion primer; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D5/08 sweep family 01 - zinc phosphate inhibitor package	Google Patents public family screen: C09D5/08 sweep family 01 - zinc phosphate inhibitor package	zinc phosphate anticorrosive pigment package; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D5/08 sweep family 02 - zinc molybdate/phosphate inhibitor blend	Google Patents public family screen: C09D5/08 sweep family 02 - zinc molybdate/phosphate inhibitor blend	mixed molybdate/phosphate inhibitor blend; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D5/08 sweep family 03 - rare-earth inhibitor epoxy coating	Google Patents public family screen: C09D5/08 sweep family 03 - rare-earth inhibitor epoxy coating	rare-earth anticorrosive pigment; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D5/08 sweep family 04 - azole inhibitor epoxy-polyamide coating	Google Patents public family screen: C09D5/08 sweep family 04 - azole inhibitor epoxy-polyamide coating	azole inhibitor in epoxy-polyamide primer; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D7/61 sweep family 01 - micaceous iron oxide barrier pigment	Google Patents public family screen: C09D7/61 sweep family 01 - micaceous iron oxide barrier pigment	MIO lamellar barrier pigment; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D7/61 sweep family 02 - aluminum flake epoxy barrier pigment	Google Patents public family screen: C09D7/61 sweep family 02 - aluminum flake epoxy barrier pigment	aluminum flake lamellar barrier pigment; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D7/61 sweep family 03 - graphene oxide epoxy barrier	Google Patents public family screen: C09D7/61 sweep family 03 - graphene oxide epoxy barrier	graphene oxide barrier nanofiller; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C09D7/61 sweep family 04 - nanoclay epoxy barrier coating	Google Patents public family screen: C09D7/61 sweep family 04 - nanoclay epoxy barrier coating	organoclay/nanoclay barrier pigment; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C08G59/50 sweep family 01 - polyamide curing agent	Google Patents public family screen: C08G59/50 sweep family 01 - polyamide curing agent	polyamide curing agent composition; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only

C08G59/50 sweep family 02 - polyamidoamine curing agent	Google Patents public family screen: C08G59/50 sweep family 02 - polyamidoamine curing agent	polyamidoamine curing agent composition; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
C08G59/50 sweep family 03 - modified amine adduct	Google Patents public family screen: C08G59/50 sweep family 03 - modified amine adduct	modified amine adduct for coatings; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
B05D sweep family 01 - high-solids epoxy spray process	Google Patents public family screen: B05D sweep family 01 - high-solids epoxy spray process	application/process window for high-solids epoxy; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
B05D sweep family 02 - surface-prep and coating system process	Google Patents public family screen: B05D sweep family 02 - surface-prep and coating system process	pretreatment and primer application process; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
B05D sweep family 03 - automated coating QC process	Google Patents public family screen: B05D sweep family 03 - automated coating QC process	process/QC monitoring for coating application; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
reactive diluent sweep family	Google Patents public family screen: reactive diluent sweep family	reactive diluent high-solids epoxy; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
self-healing microcapsule sweep family	Google Patents public family screen: self-healing microcapsule sweep family	microcapsule self-healing epoxy coating; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
intrinsic self-healing epoxy sweep family	Google Patents public family screen: intrinsic self-healing epoxy sweep family	dynamic-bond self-healing epoxy matrix; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
waterborne epoxy anticorrosive sweep family	Google Patents public family screen: waterborne epoxy anticorrosive sweep family	waterborne epoxy anticorrosive coating; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
zinc-rich epoxy primer sweep family	Google Patents public family screen: zinc-rich epoxy primer sweep family	zinc-rich epoxy cathodic-protection primer; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only
process data/model optimization sweep family	Google Patents public family screen: process data/model optimization sweep family	formulation optimization and coating process data loop; public claim family to be selected by counsel; baseline overlap unresolved	CPC/IPC sweep row: no specific independent claim has been counsel-charted to a clearance conclusion.	UNCLEAR / not FTO / category/family screen; individual family status unverified / not FTO; category/family screen only

12. Molecule cards

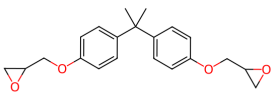
The dedicated MOLECULE_ENGINE.pdf remains the specialized molecule-engine appendix; representative cards are included here for bundle completeness.

13. Molecule Cards — structure/property appendix (known vs novel)

Each card shows the chemical structure, the model's property, and a KNOWN-in-industry vs NOVEL badge: **KNOWN / INDUSTRY STANDARD** **KNOWN / COMPOUND** **EMERGING / COMMERCIAL** **NOVEL / RESEARCH**.

Epoxy resins

#1 Bisphenol-A diglycidyl ether (DGEBA) — KNOWN / INDUSTRY STANDARD



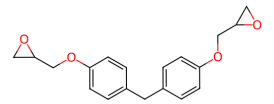
rendered from SMILES

Role	resin
EEW computed	170.2 g/eq (monomer, exact)
EEW measured	188.5 g/eq (commercial, +oligomer)
Functionality	2
Viscosity	130 P
Tier	COMPUTED + MEASURED
Spec fit	1.00

Verified: commercial grade + supplier TDS PubChem CID 2286

the workhorse liquid epoxy of MIL-DTL-24441 primers

#2 Bisphenol-F diglycidyl ether (DGEBF) — KNOWN / INDUSTRY STANDARD



rendered from SMILES

Role	resin
EEW computed	156.2 g/eq (monomer, exact)
EEW measured	169.0 g/eq (commercial, +oligomer)
Functionality	2
Viscosity	35 P
Tier	COMPUTED + MEASURED
Spec fit	0.86

Verified: commercial grade + supplier TDS PubChem CID 91511

lower-viscosity BPF analog; helps hit 65% solids sprayable

#3 Epoxy phenol-novolac (D.E.N. 438 / EPON 154) — KNOWN / INDUSTRY STANDARD

No single canonical structure (polydisperse / oligomeric commercial product). Equivalent weight is a MEASURED distribution average (TDS titration), not computed.

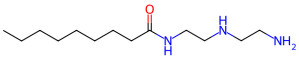
Role	resin
EEW measured	178.5 g/eq (TDS, polydisperse)
Functionality	3.6
Tier	MEASURED TDS
Spec fit	0.84

Verified: commercial grade + supplier TDS CAS 28064-14-4 + supplier TDS

high-functionality novolac; chemical resistance, higher crosslink – discovery-adjacent

Polyamide curatives (the envelope chemistry)

#1 Ancamide 2445 (modified polyamide, marine) — KNOWN / INDUSTRY STANDARD

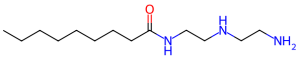


representative motif — product is a polydisperse oligomer distribution

Role	hardener
AHEW	150 g/eq (MEASURED, titrated)
Class	reactive polyamide
Tier	MEASURED
Spec fit	0.94

Verified: commercial grade + supplier TDS supplier product guide (measured TDS)
high-solids industrial-maintenance & MARINE polyamide

#2 Ancamide 2573 (modified polyamide) — KNOWN / INDUSTRY STANDARD

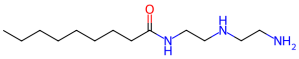


representative motif — product is a polydisperse oligomer distribution

Role	hardener
AHEW	115 g/eq (MEASURED, titrated)
Class	reactive polyamide
Tier	MEASURED
Spec fit	0.94

Verified: commercial grade + supplier TDS supplier product guide (measured TDS)
surface-tolerant primer, industrial/marine

#3 Ancamide 260A (dimer-acid polyamide) — KNOWN / INDUSTRY STANDARD



representative motif — product is a polydisperse oligomer distribution

Role	hardener
AHEW	110 g/eq (MEASURED, titrated)
Class	reactive polyamide
Tier	MEASURED
Spec fit	0.90

Verified: commercial grade + supplier TDS supplier product guide (measured TDS)
industry-standard dimer-acid polyamide

Discovery Annex — novel / emerging epoxies

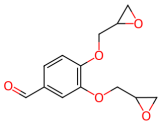
NEW Cardanol (CNSL) glycidyl ether (Cardolite NC-514 — EMERGING / COMMERCIAL

No single canonical structure (polydisperse / oligomeric commercial product). Equivalent weight is a MEASURED distribution average (TDS titration), not computed.

Class	novel / emerging epoxy
EEW	[400, 500] g/eq (measured, emerging product)
Tier	**MEASURED (emerging)**

Verified: on-market bio-based product MDPI Polymers 17(14):1956 (2025) review; Cardolite
bio-based flexibilizing epoxy; C8+ side chain hydrophobicity

NEW Diglycidyl vanillin epoxy — NOVEL / RESEARCH

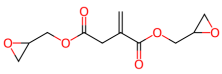


rendered from SMILES

Class	novel / emerging epoxy
EEW (predicted)	125.1 g/eq (structure, IDEAL-monomer)
Tier	**STRUCTURE-COMPUTED (unmade)**

Verified: research-stage, unmade Eur. Polym. J. S1381514821002261
bio-based, flame-retardant + degradable; EEW computable from structure

NEW Diglycidyl ester of itaconic acid — NOVEL / RESEARCH



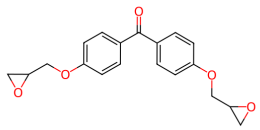
rendered from SMILES

Class	novel / emerging epoxy
EEW (predicted)	121.1 g/eq (structure, IDEAL-monomer)
Tier	**STRUCTURE-COMPUTED (unmade)**

Verified: research-stage, unmade MDPI Polymers 17(14):1956 (2025) review
bio-platform diacid epoxy

GENERATED de-novo candidates (engine-designed; backbone Bisphenol A regenerates DGEBA)

#1 Glycidyl ether of 4,4'-Dihydroxybenzophenone — GENERATED / DE-NOVO

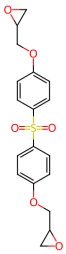


rendered from SMILES

Backbone	4,4'-Dihydroxybenzophenone
EEW (computed)	163.2 g/eq (structure, exact)
Functionality	2
Tier	GENERATED (unmade, structure-computed)
Design score	1.00

Basis: de-novo design lead
rigid ketone-linked, non-BPA

#2 Glycidyl ether of Bisphenol S (sulfone) — GENERATED / DE-NOVO

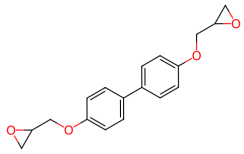


rendered from SMILES

Backbone	Bisphenol S (sulfone)
EEW (computed)	181.2 g/eq (structure, exact)
Functionality	2
Tier	GENERATED (unmade, structure-computed)
Design score	1.00

Basis: de-novo design lead
non-BPA rigid sulfone; higher Tg/flame resistance

#3 Glycidyl ether of 4,4'-Biphenol — GENERATED / DE-NOVO



rendered from SMILES

Backbone	4,4'-Biphenol
EEW (computed)	149.2 g/eq (structure, exact)
Functionality	2
Tier	GENERATED (unmade, structure-computed)
Design score	1.00

Basis: de-novo design lead
rigid biphenyl; high modulus, non-BPA

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